Capstone hypothesis and baseline

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

One readable capstone planning record.

Keep the hypothesis, fixed choices, baseline arithmetic, reserved roles and unrun status together.

Keep one habit: Recall → W1–W3/W5 → independent plan → project version → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Documenting a network experiment. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Dataset shift in neural models. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Similarity and normalisation. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- A capstone needs a question that **evidence** can answer. “Build a better network” leaves better undefined.

  Here the question is: does two-update full-batch **fitting** of one scalar ReLU weight reduce validation mean squared error relative to the initial-weight **baseline** on a constructed positive-**input** task?

  Predict max(0, a*x), with no bias. The **baseline** has a = 1. The trained candidate starts at a = 1, uses learning rate 0.1 and exactly two updates. These choices are written before results are inspected.

- The **hypothesis** is that the trained candidate has lower validation MSE. The plan does not guarantee that result. This task tests **training** and record keeping, not whether neural networks are preferable to every other model.

  A stronger comparison may also include an ordinary fitted line.

Documenting a network experiment

> **My remembered idea, example and correction**
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Dataset shift in neural models

> **My remembered idea, example and correction**

Similarity and normalisation

> **My remembered idea, example and correction**

Lesson reference for workbook practice

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main plan CAP33: prediction ReLU(a*x), no bias; baseline a1; candidate starts a1, learning rate .1, exactly two full-batch MSE updates. Training T1 (1, 2), T2 (2, 4). Validation V1 (3, 6), V2 (4, 8). Baseline predictions 3/4, errors −3/−4, squared errors 9/16, MSE 12.5 on two rows. Strict validation criterion: candidate MSE <12.5 on those same rows; equality fails.

  Candidate result not yet run; final result not measured. Final cases remain outside fitting, tuning and selection; record actual exposure and reserve fresh cases if already seen. Separate supported baseline predictions 2/4 versus targets 5/7 gives errors −3/−3 and MSE 9, but candidate predictions missing.

  Separate picture proposes corner-only A versus centre-feature B on eight reserved patterns; it supplies no scored predictions or exact model/data fixture. It teaches a planning distinction, not a measured improvement. A revised learning-rate plan needs a preserved version and an honest account of what informed it.

- State the **prediction** rule, **baseline**, candidate learning rate and update count, comparison data and metric. Write a testable **hypothesis**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main plan CAP33: prediction ReLU(a*x), no bias; baseline a1; candidate starts a1, learning rate .1, exactly two full-batch MSE updates. Training T1 (1, 2), T2 (2, 4). Validation V1 (3, 6), V2 (4, 8). Baseline predictions 3/4, errors −3/−4, squared errors 9/16, MSE 12.5 on two rows. Strict validation criterion: candidate MSE <12.5 on those same rows; equality fails.

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  Separate picture proposes corner-only A versus centre-feature B on eight reserved patterns; it supplies no scored predictions or exact model/data fixture. It teaches a planning distinction, not a measured improvement. A revised learning-rate plan needs a preserved version and an honest account of what informed it.

- Compute **baseline prediction**s, errors, squared errors and MSE on V1 and V2. State the denominator.

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main plan CAP33: prediction ReLU(a*x), no bias; baseline a1; candidate starts a1, learning rate .1, exactly two full-batch MSE updates. Training T1 (1, 2), T2 (2, 4). Validation V1 (3, 6), V2 (4, 8). Baseline predictions 3/4, errors −3/−4, squared errors 9/16, MSE 12.5 on two rows. Strict validation criterion: candidate MSE <12.5 on those same rows; equality fails.

  Candidate result not yet run; final result not measured. Final cases remain outside fitting, tuning and selection; record actual exposure and reserve fresh cases if already seen. Separate supported baseline predictions 2/4 versus targets 5/7 gives errors −3/−3 and MSE 9, but candidate predictions missing.

  Separate picture proposes corner-only A versus centre-feature B on eight reserved patterns; it supplies no scored predictions or exact model/data fixture. It teaches a planning distinction, not a measured improvement. A revised learning-rate plan needs a preserved version and an honest account of what informed it.

- A proposal says “try models until the test score looks good”. Identify the missing fixed choices and explain why that test is not independent.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main plan CAP33: prediction ReLU(a*x), no bias; baseline a1; candidate starts a1, learning rate .1, exactly two full-batch MSE updates. Training T1 (1, 2), T2 (2, 4). Validation V1 (3, 6), V2 (4, 8). Baseline predictions 3/4, errors −3/−4, squared errors 9/16, MSE 12.5 on two rows. Strict validation criterion: candidate MSE <12.5 on those same rows; equality fails.

  Candidate result not yet run; final result not measured. Final cases remain outside fitting, tuning and selection; record actual exposure and reserve fresh cases if already seen. Separate supported baseline predictions 2/4 versus targets 5/7 gives errors −3/−3 and MSE 9, but candidate predictions missing.

  Separate picture proposes corner-only A versus centre-feature B on eight reserved patterns; it supplies no scored predictions or exact model/data fixture. It teaches a planning distinction, not a measured improvement. A revised learning-rate plan needs a preserved version and an honest account of what informed it.

- A partner inspects final labels and changes the learning rate. Explain how to document this and obtain independent **evidence**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A new task predicts max(0, a*x), with **baseline** a = 1. **Training** rows are (1,3) and (2,6); validation rows are (3,9) and (4,12); final rows are sealed.

  Specify a candidate starting at 1 with learning rate 0.1 and two full-batch updates. Calculate **baseline** validation MSE and write a strict improvement criterion. Has the candidate succeeded yet?

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Plan a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the mathematical evidence

> Keep this task's inputs,rule and response together.

Complete task and evidence

- For Image-pattern Network Investigation, use this week's independent task as a rehearsal for the final **evidence** pack: A new task predicts max(0, a*x), with **baseline** a = 1.

  Training rows are (1,3) and (2,6); validation rows are (3,9) and (4,12); final rows are sealed. Specify a candidate starting at 1 with learning rate 0.1 and two full-batch updates.

  Calculate **baseline** validation MSE and write a strict improvement criterion. Has the candidate succeeded yet?

  Keep actual **observation**s separate from expected results, preserve failures, compare a **baseline** and explain the limits.

> **My project change, supporting evidence and remaining limit**
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MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	

Evidence	My record
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Calculate the reference score for Capstone **hypothesis** and **baseline**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Capstone **hypothesis** and **baseline**. Proposed explanation: "A **hypothesis** is a result". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A new task predicts max(0, a*x), with **baseline** a = 1. **Training** rows are (1,3) and (2,6); validation rows are (3,9) and (4,12); final rows are sealed.

  Specify a candidate starting at 1 with learning rate 0.1 and two full-batch updates. Calculate **baseline** validation MSE and write a strict improvement criterion. Has the candidate succeeded yet?

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A testable image-pattern **hypothesis**. A team proposes that including the centre-pixel feature improves a small fixed network's classification of toy image patterns.

- Two frozen versions are planned: A uses corner features only; B also uses the centre feature. Eight reserved toy patterns have labels defined before scoring.

- Name the intended change, the **baseline** and the measured outcome. Why must both versions avoid **training** on the eight reserved labels?

- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record.

  State exactly what this example supports before linking it to the wider capstone claim.

AI ACADEMY / AI-5/33

Capstone hypothesis and baseline

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Write the prediction

B should have more correct labels than A on the same eight patterns.



2. Declare the baseline

A is a documented corner-only network, not a deliberately broken version.



3. Reserve evaluation

Keep the eight patterns out of model selection and tuning.

CHANGE ONE THING

Give only B the eight evaluation labels during training.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Image-pattern Network Investigation
- Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.
- For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- A testable image-pattern **hypothesis**. A team proposes that including the centre-pixel feature improves a small fixed network's classification of toy image patterns.
- Which labelled step matches this detail? Keep the eight patterns out of model selection and tuning.
- 1. Reserve evaluation
- 2. Declare the **baseline**
- 3. Write the **prediction**
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Reserve evaluation Keep the eight patterns out of model selection and tuning. A capstone **hypothesis** links a specific architectural change to a measured outcome.

A **baseline** anchors the comparison; frozen labels and held-out patterns protect it from answer-key tuning.

Changed-case question and optional hint

- Changed-case practice: Give only B the eight evaluation labels during **training**. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. The proposed comparison becomes contaminated rather than a valid held-out test.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The proposed comparison becomes contaminated rather than a valid held-out test. One version has gained privileged information from the reserved evaluation set.

Repair and retain the record

Review Documenting a network experiment (ai-5/32). Goal: Record a network experiment with enough detail to replay its calculations. Short practice: What additional **evidence** is needed for a final performance claim?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Dataset shift in neural models (ai-5/30). Goal: Explain how a changed **input** mix can change overall accuracy. Short practice: Did the supplied model's dim-light accuracy deteriorate?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Similarity and normalisation (ai-5/27). Goal: Calculate vector length and normalise a nonzero vector. Short practice: Why do the original two metrics select different winners?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Capstone hypothesis and baseline: the case

AI ACADEMY / AI-5/33 · 01 · LOOK AT THE FACTS

Capstone hypothesis and baseline: the case

The situation

A testable image-pattern hypothesis. A team proposes that including the centre-pixel feature improves a small fixed network's classification of toy image patterns.

Supplied evidence

Two frozen versions are planned: A uses corner features only; B also uses the centre feature. Eight reserved toy patterns have labels defined before scoring.

Explain capstone hypothesis and baseline with a traceable toy-network case and a bounded conclusion.

Follow the labelled picture with the matching explanation. Open the vector diagram

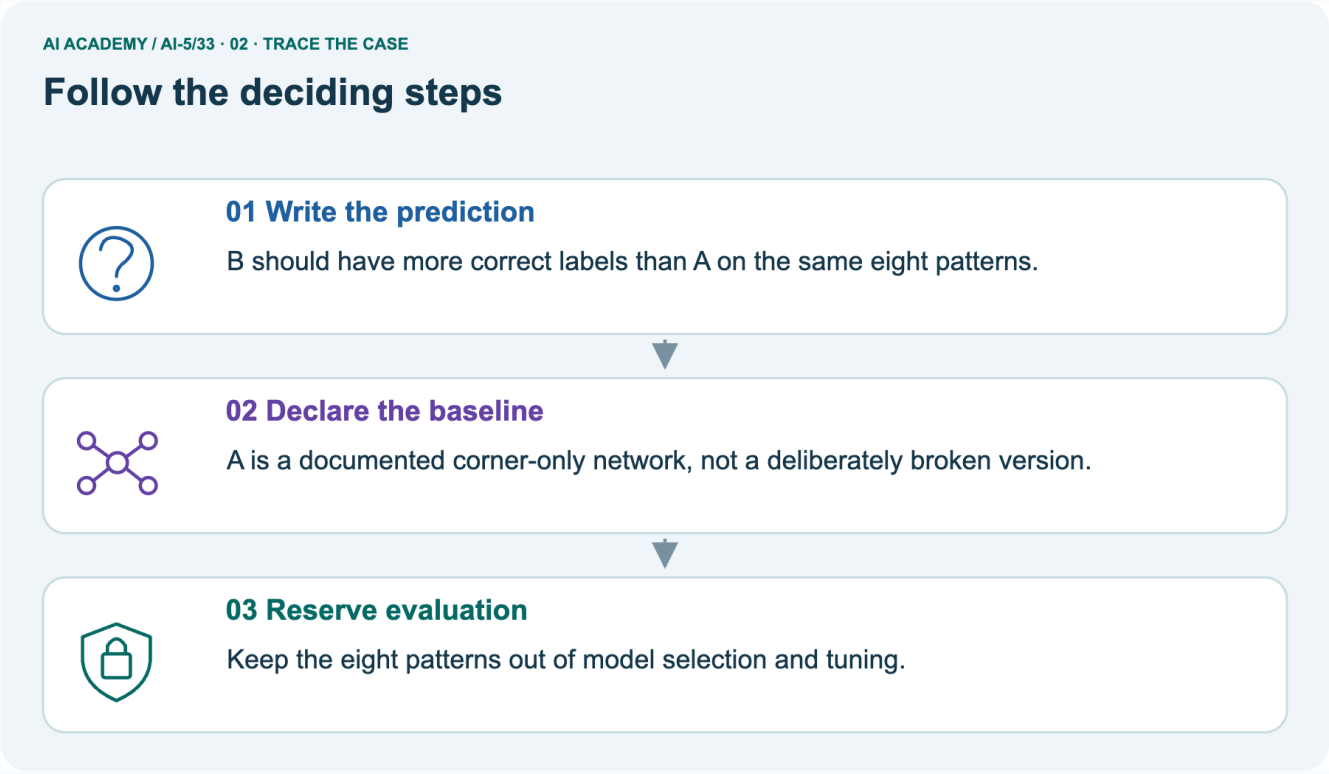
A testable image-pattern **hypothesis**. A team proposes that including the centre-pixel feature improves a small fixed network's classification of toy image patterns.

Supplied facts: Two frozen versions are planned: A uses corner features only; B also uses the centre feature. Eight reserved toy patterns have labels defined before scoring.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



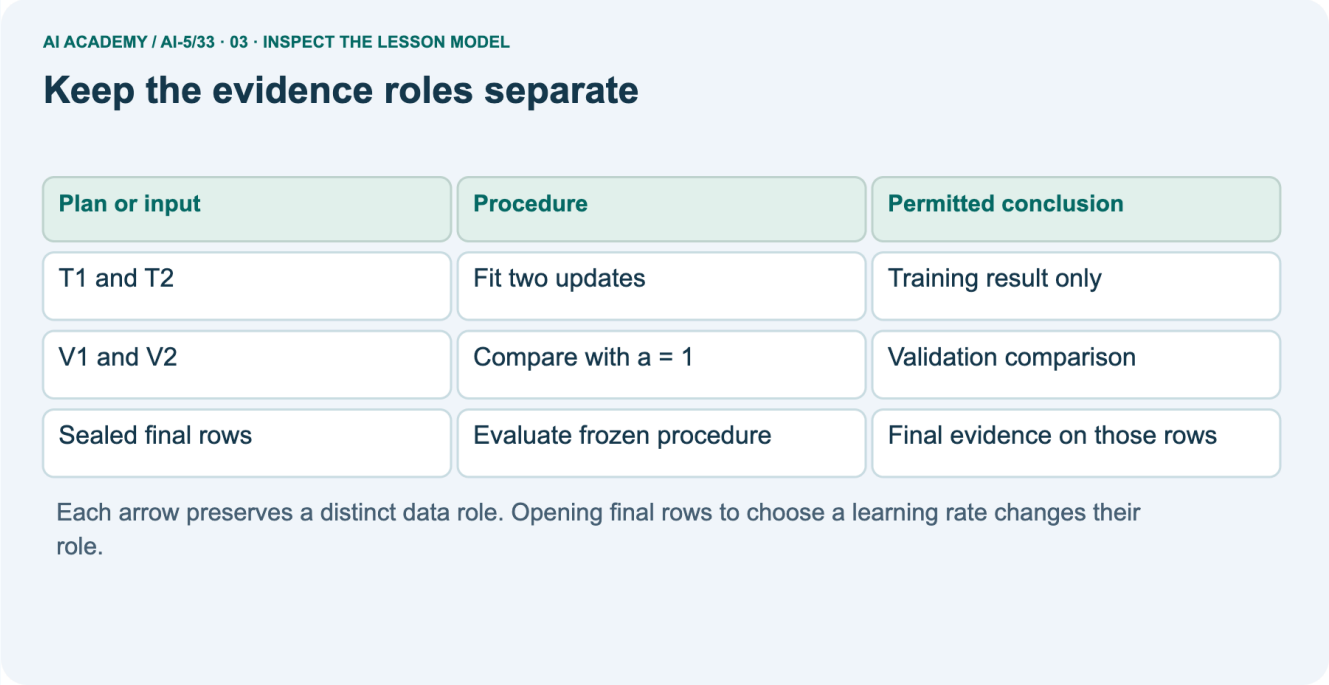
Follow the labelled picture with the matching explanation. Open the vector diagram

Write the **prediction**: B should have more correct labels than A on the same eight patterns. → Declare the **baseline**: A is a documented corner-only network, not a deliberately broken version. → Reserve evaluation: Keep the eight patterns out of model selection and tuning.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Keep the evidence roles separate



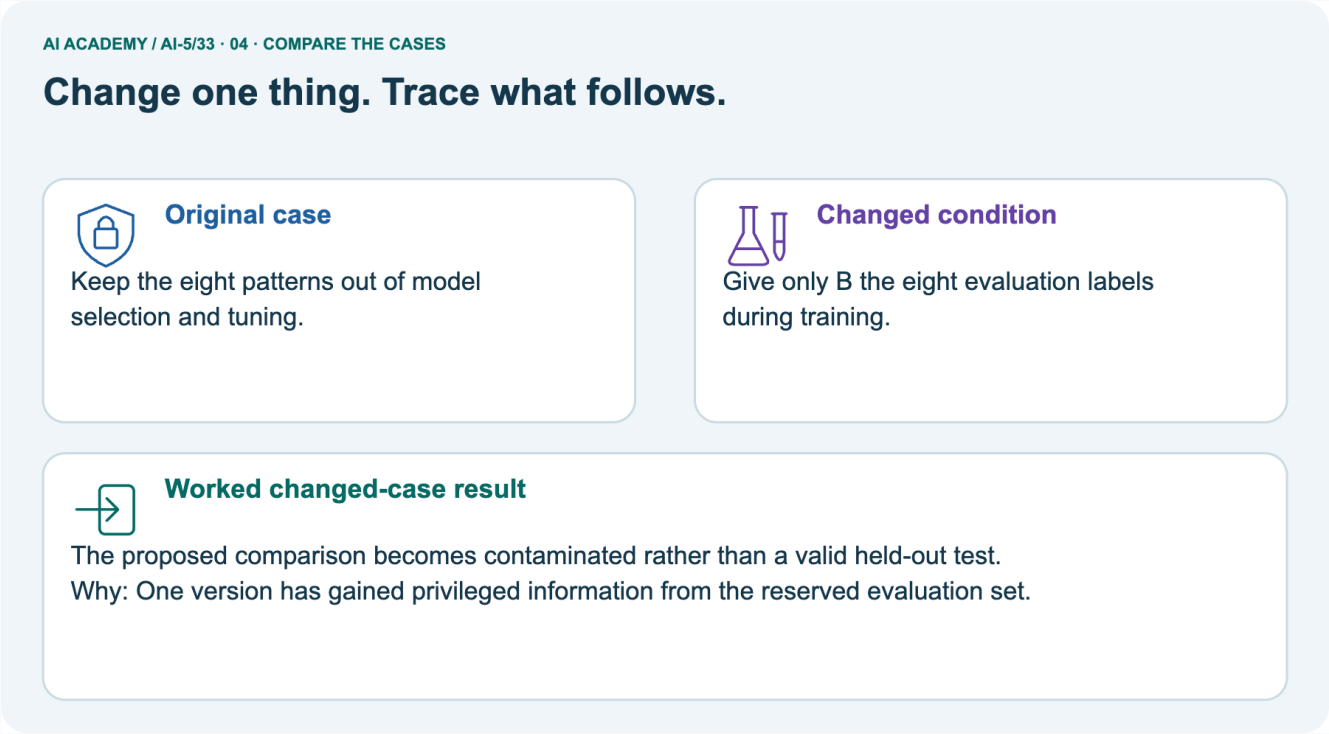
Follow the labelled picture with the matching explanation. Open the vector diagram

Keep the **evidence** roles separate. T1 and T2 | Fit two updates | **Training** result only. V1 and V2 | Compare with a = 1 | Validation comparison. Sealed final rows | Evaluate frozen procedure | Final **evidence** on those rows

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

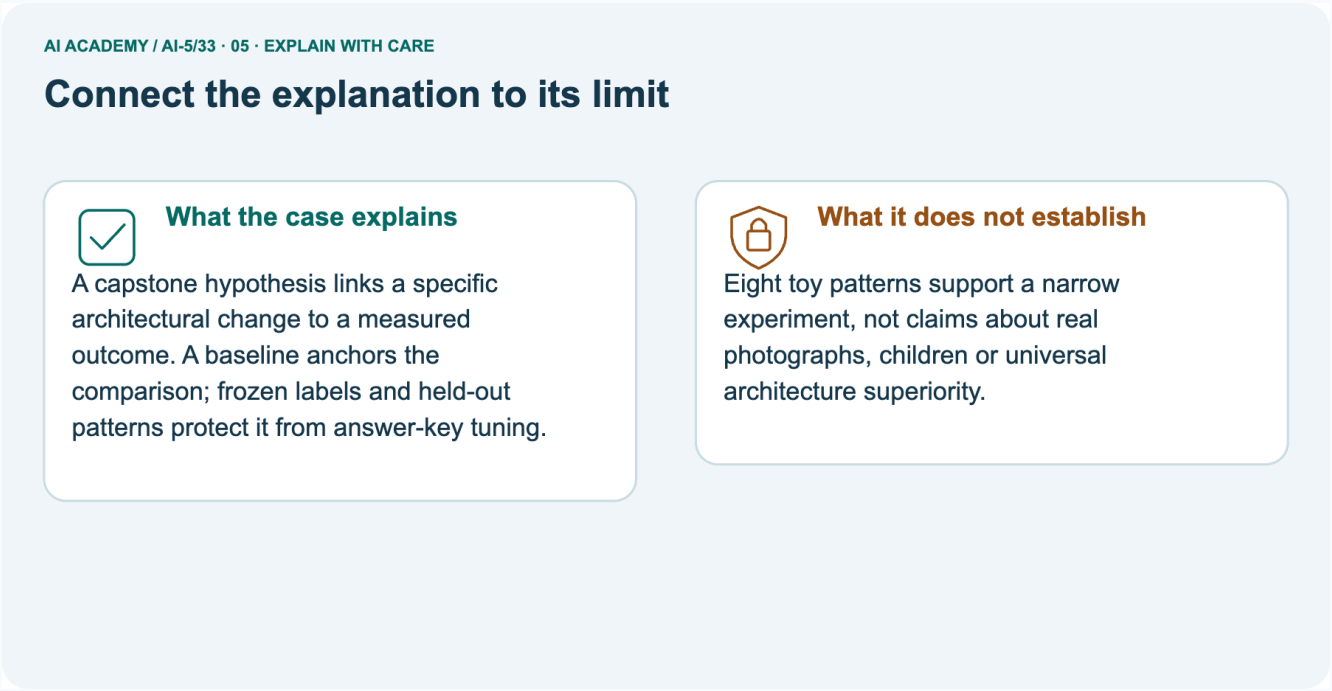
Original case: Keep the eight patterns out of model selection and tuning. Changed condition: Give only B the eight evaluation labels during **training**.

Expected result: The proposed comparison becomes contaminated rather than a valid held-out test. Reason: One version has gained privileged information from the reserved evaluation set.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

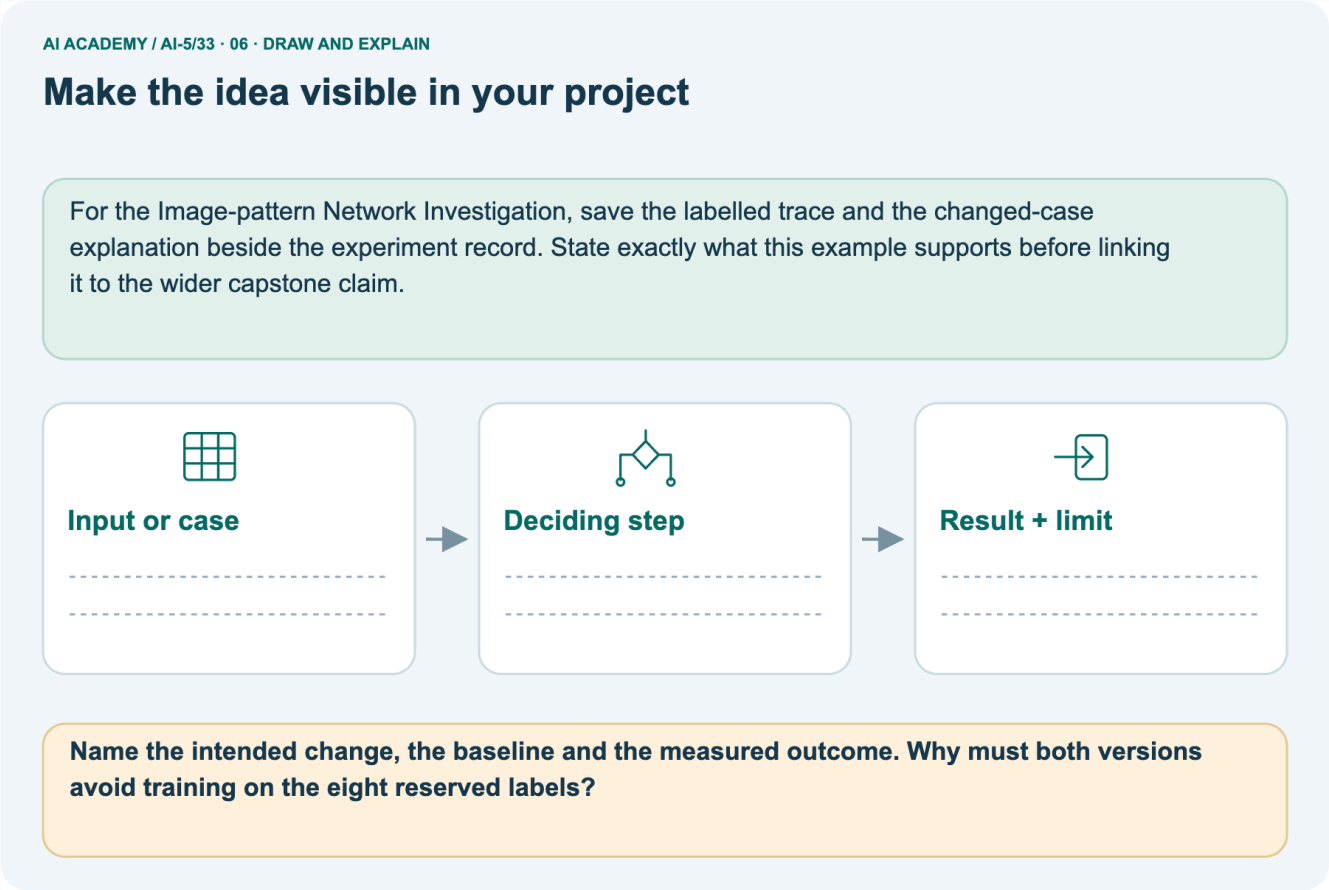
A capstone **hypothesis** links a specific architectural change to a measured outcome. A **baseline** anchors the comparison; frozen labels and held-out patterns protect it from answer-key tuning.

Limit: Eight toy patterns support a narrow experiment, not claims about real photographs, children or universal architecture superiority.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

Draw your **input**, deciding step, result and limit. Supported question: Name the intended change, the **baseline** and the measured outcome. Why must both versions avoid **training** on the eight reserved labels?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Capstone hypothesis and baseline: the case

02. Follow the deciding steps

03. Keep the evidence roles separate

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.